A TRANSPOSITION-FIRST MANUAL

Piano Without Fear of Black Keys

A transposition-first manual for piano, Ableton Push/ Note, and guitar

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Preface

This is not a general piano method. It has a narrower goal: make **scales, chords, arpeggios, voicings, and progressions transposable** on a keyboard whose physical geometry changes from key to key. It starts from first principles so a new musician can use it, but everything points back to that goal.

You do **not** need to read staff notation, have perfect pitch, memorize every key signature, or already know piano fingering. You need a keyboard or piano, a few minutes of focused practice, and the willingness to say note names and interval formulas before speed becomes important.

How to read every musical example

The book uses the same four-layer translation throughout:

```
function → pitch spelling → register → instrument geometry
```

For example, a minor seventh chord is always:

```
1 b3 5 b7
```

On D, that becomes:

```
D F A C
```

Three geometries for the same music

4. Piano: non-isomorphic geometry

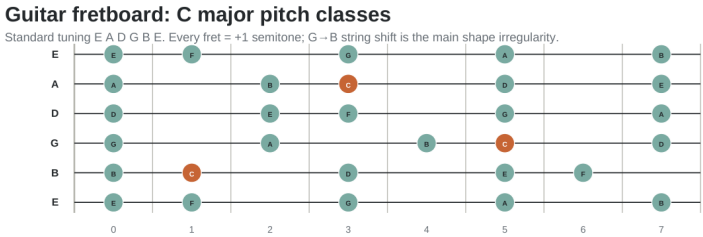
On piano, the same interval formula produces different visible shapes in different keys. Compare:

```
Cmaj7      Dbmaj7
```

Both are `1 3 5 7`. The musical object is identical; only its keyboard realization changes.

7. Guitar: mostly movable geometry

In standard tuning, every fret is a semitone. Most interval/chord shapes transpose by sliding, with one important irregularity: the interval between G and B strings is a major third rather than a perfect fourth.



Guitar C major

The piano problem is therefore the reverse of a barre chord: the *formula* transposes, but the *shape* does not.

7.1 Movable shapes hide pitch location

Movable geometry is useful, but it creates a specific blind spot. On guitar, a scale or chord can often be transposed by moving a familiar shape up or down the neck. The fingering relationship survives, but every sounding pitch has moved with it. A player can therefore know *how to move the object* without knowing which notes are sounding, or where the same pitch classes live in another register. Push and Note can encourage the same habit because their chromatic layouts are largely isomorphic.

Piano exposes the problem differently. Several keys can be practised in the same register: C major around middle C, Db major beginning a semitone higher, D major another semitone higher. The hand geometry changes, but the keyboard remains a fixed map of absolute pitches. This makes it easier to notice that transposition and relocation are not the same operation.

The book therefore treats **shape as a consequence, not as the musical object**. A shape is allowed to become automatic only after you can recover what it means: function, note names, register, and exact locations.

> **Fixed-register rule.** When practising transposition on guitar, Push, or Note, do not always slide the same shape. Choose a register window first —for example C3–C5—and realize each scale or chord inside that window. Changing key will then force new strings, pads, inversions, or fingerings while keeping the sounding range comparable.

9. The seven diatonic modes

The modes are not seven unrelated scales. They are seven rotations / alterations of a diatonic collection, best understood as degree formulas relative to a new tonic.

Mode	Formula	Characteristic degree
Ionian	`1 2 3 4 5 6 7`	major 7
Dorian	`1 2 b3 4 5 6 b7`	natural 6 over minor
Phrygian	`1 b2 b3 4 5 b6 b7`	b2
Lydian	`1 2 3 #4 5 6 7`	#4
Mixolydian	`1 2 3 4 5 6 b7`	b7 over major
Aeolian	`1 2 b3 4 5 b6 b7`	b6
Locrian	`1 b2 b3 4 b5 b6 b7`	b5

Exercise 5 - Break the white-key trap

If C major, A minor, and D Dorian are your default universe, they all use the same pitch collection. Deliberately practice modes whose tonic forces black keys:

- Eb Dorian: `Eb F Gb Ab Bb C Db`
- F# Dorian: `F# G# A B C# D# E`
- Db Lydian: `Db Eb F G Ab Bb C`
- Ab Mixolydian: `Ab Bb C Db Eb F Gb`

For each, alternate between:

```
scale → triad → seventh chord → arpeggio → 2-bar
improvisation
```

Chords, inversions, voicings, and arpeggios

11. Chords are formulas

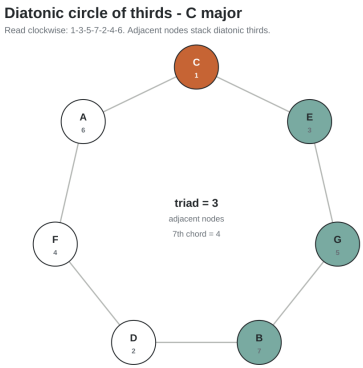
Read a chord symbol from left to right: **root letter → chord quality → extensions or alterations**. For example, `Abmaj9` means an Ab-root chord with major quality, a major seventh, and a ninth.

11.1 The circle of thirds: the geometry inside tertian chords

In a diatonic key, reorder the seven scale degrees as:

`1 → 3 → 5 → 7 → 2 → 4 → 6 → 1`

In C major this is `C → E → G → B → D → F → A → C`. Three adjacent positions give a triad; four give a seventh chord. Unlike the circle of fifths, this **diatonic circle of thirds depends on the key** because the sequence uses scale notes rather than a fixed chromatic interval.



Diatonic circle of thirds in C major